

Introduction to Data Science — Final Exam Paper 3 (Easy)

Course Code: BCSE2C05 | Semester: II | Max Marks: 100 | Time: 3 Hours Difficulty: ★☆☆☆☆ Easy

SECTION A — Short Answer (5 Marks Each, Do Any 4 of 5 = 20 Marks)

Q1. What is Data Mining? Differentiate between Descriptive and Predictive Data Mining.

Data Mining is the process of discovering patterns, correlations, and useful knowledge from large datasets using statistical and computational techniques.

Aspect	Descriptive Mining	Predictive Mining
Purpose	Summarizes past data ("what happened").	Predicts future outcomes ("what will happen").
Techniques	Clustering, Association, Summarization	Classification, Regression, Forecasting
Example	"70% of buyers aged 20-30 prefer online shopping."	"Customer X has 85% chance of buying product Y."

Q2. What is a Pandas DataFrame? How do you create one from a dictionary?

A **DataFrame** is a 2D, labelled data structure in Pandas — like a spreadsheet with rows and columns.

```
import pandas as pd

data = {
    'Name': ['Rajat', 'Priya', 'Aman'],
    'Age': [21, 22, 20],
    'CGPA': [8.5, 9.0, 7.8]
}
df = pd.DataFrame(data)
print(df)
```

Output:

	Name	Age	CGPA
0	Rajat	21	8.5
1	Priya	22	9.0
2	Aman	20	7.8

Q3. Explain Range, Quartiles, and Interquartile Range (IQR) with an example.

Given data: {5, 10, 15, 20, 25, 30, 35}

Measure	Definition	Calculation	Result
Range	Max – Min	35 – 5	30
Q1	25th percentile (median of lower half)	Median of {5, 10, 15}	10
Q2	50th percentile (median)	Middle value	20
Q3	75th percentile (median of upper half)	Median of {25, 30, 35}	30
IQR	Q3 – Q1	30 – 10	20

IQR is useful for detecting outliers: values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are outliers.

Q4. What is Seaborn? How is it different from Matplotlib?

Seaborn is a Python visualization library built on top of Matplotlib that provides a high-level interface for creating attractive statistical graphics.

Feature	Matplotlib	Seaborn
Level	Low-level (more control)	High-level (easier syntax)
Default Style	Basic, plain	Beautiful, modern themes
Statistical Plots	Manual effort needed	Built-in (boxplot, heatmap, violin)
DataFrames	Not natively supported	Works directly with Pandas DataFrames
Code Required	More lines	Fewer lines for same plot

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset('tips')
sns.boxplot(x='day', y='total_bill', data=tips)
plt.title('Bill Distribution by Day')
plt.show()
```

Q5. List the roles in a Data Science team and briefly explain each.

Role	Responsibility
Data Scientist	Analyses data, builds ML models, extracts insights.
Data Engineer	Designs and maintains data pipelines and infrastructure.
Data Analyst	Explores data, creates reports and dashboards.
ML Engineer	Deploys and scales ML models into production.
Business Analyst	Translates business needs into data requirements.
Domain Expert	Provides subject-matter knowledge for proper interpretation.

SECTION B — Analytical Questions (8 Marks Each, Do Any 5 of 6 = 40 Marks)

Q6. Explain Data Collection Strategies. What are the different ways to collect data?

Data Collection is the systematic process of gathering information from relevant sources.

Strategies:

Strategy	Description	Example
Surveys/Questionnaires	Collecting responses from participants.	Google Forms survey on customer satisfaction.
Interviews	One-on-one structured conversations.	Interviewing doctors for medical research.
Observation	Watching and recording behaviour.	Tracking user clicks on a website.
Experiments	Controlled tests to gather data.	A/B testing on a landing page.
Web Scraping	Extracting data from websites programmatically.	Scraping product prices from e-commerce sites.
APIs	Accessing data through application interfaces.	Twitter API for tweet data.
Existing Databases	Using pre-existing records.	Census data, hospital records.

Sensor/IoT Data	Automated data collection from devices.	Temperature sensors in a factory.
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Q7. What is Data Integration and Data Transformation? Explain with examples.

Data Integration: Combining data from multiple sources into a unified view.

- **Challenge:** Data from different sources may have different formats, schemas, or naming conventions.
- **Example:** Merging customer data from a CRM system (has Customer_ID, Name, Email) with sales data (has Cust_ID, Product, Amount). Requires matching Customer_ID with Cust_ID.

Techniques: Entity resolution, schema matching, duplicate detection.

Data Transformation: Converting data from one format or structure to another to make it suitable for analysis.

Technique	Description	Example
Normalization	Scale values to [0,1] range.	Min-Max scaling: (X-min)/(max-min)
Aggregation	Summarize data at a higher level.	Daily sales → Monthly totals
Encoding	Convert categorical to numerical.	Male → 1, Female → 0 (label encoding)
Discretization	Convert continuous to bins.	Age: 0-18=Child, 19-60=Adult, 60+=Senior

Q8. Explain forward fill, backward fill, and interpolation for handling missing values with code.

```
import pandas as pd
import numpy as np

df = pd.DataFrame({
    'Day': [1, 2, 3, 4, 5, 6],
    'Temp': [30, np.nan, np.nan, 36, np.nan, 40]
})

# Forward Fill – uses previous value
df['ffill'] = df['Temp'].ffill()
# Result: 30, 30, 30, 36, 36, 40

# Backward Fill – uses next value
df['bfill'] = df['Temp'].bfill()
# Result: 30, 36, 36, 36, 40, 40

# Linear Interpolation – estimates value linearly
df['interpolated'] = df['Temp'].interpolate(method='linear')
# Result: 30, 32, 34, 36, 38, 40

print(df)
```

Day	Temp	ffill	bfill	interpolated
1	30	30	30	30
2	NaN	30	36	32
3	NaN	30	36	34
4	36	36	36	36
5	NaN	36	40	38
6	40	40	40	40

Q9. Explain Symmetric and Asymmetric Binary Attributes with examples.

Binary Attributes have only two possible values (0 or 1, True or False).

Symmetric Binary:

- Both values (0 and 1) are **equally important**.
- Example: **Gender** (Male = 0, Female = 1) — both values are equally significant.
- When computing similarity, both 0-0 and 1-1 matches are counted.

Asymmetric Binary:

- One value is **more important or rare** than the other.
- The rare outcome (usually 1) is of primary interest.
- Example: **HIV Test Result** (Positive = 1, Negative = 0) — a positive result is far more significant than negative.
- When computing similarity, only 1-1 matches matter; 0-0 matches are ignored.

Jaccard Coefficient is used for asymmetric binary attributes:

J = f₁₁ / (f₀₁ + f₁₀ + f₁₁)

Where f₁₁ = both positive, f₀₁ = first 0 second 1, f₁₀ = first 1 second 0.

Q10. Write Python code to read a CSV, filter rows, and save the result to a new CSV.

```
import pandas as pd

# Read CSV
df = pd.read_csv('employees.csv')
print("Original shape:", df.shape)
print(df.head())

# Filter: employees with salary > 50000 and age > 25
filtered = df[(df['Salary'] > 50000) & (df['Age'] > 25)]
print("\nFiltered records:", len(filtered))
print(filtered.head())

# Sort by salary descending
filtered_sorted = filtered.sort_values('Salary', ascending=False)
print("\nTop 5 highest paid:")
print(filtered_sorted.head())

# Save to new CSV
filtered_sorted.to_csv('high_earners.csv', index=False)
print("\nSaved to high_earners.csv")
```

Q11. Explain the steps involved in building a Machine Learning model.

Steps for Model Building:

Step	Description	Example
1. Define Problem	Clearly state what to predict.	"Predict house prices."
2. Collect Data	Gather relevant data.	Historical house sales data.
3. Preprocess Data	Clean, transform, handle missing values.	Remove nulls, normalize prices.
4. Feature Selection	Choose relevant input features.	Area, bedrooms, location.
5. Split Data	Divide into training and testing sets.	80% train, 20% test.

6. Choose Algorithm	Select appropriate ML model.	Linear Regression for continuous output.
7. Train Model	Fit the model on training data.	<code>model.fit(X_train, y_train)</code>
8. Evaluate Model	Test on unseen data using metrics.	RMSE, R^2 , MAE on test set.
9. Tune Hyperparameters	Optimize model settings.	GridSearchCV for best parameters.
10. Deploy Model	Put into production for real use.	API endpoint for price prediction.

SECTION C — Long Answer (20 Marks Each, Do Any 2 of 3 = 40 Marks)

Q12. (a) Explain the evolution and applications of Data Science (10 Marks). (b) Describe 5 sources of data with challenges (10 Marks).

(a) Evolution of Data Science:

Era	Development
1960s	Statistics and early computing — basic data analysis.
1970s-80s	Relational databases (SQL) — structured data storage.
1990s	Data Mining emerges — discovering patterns in databases.
2000s	Big Data era — Hadoop, distributed computing.
2010s	Machine Learning & Deep Learning boom — AI-driven insights.
2020s	AI, AutoML, real-time analytics, LLMs — democratization of Data Science.

Applications:

- Healthcare** — Disease prediction, drug discovery, medical imaging.
- Finance** — Fraud detection, algorithmic trading, credit scoring.
- E-commerce** — Recommendation systems, inventory optimization.
- Transportation** — Autonomous vehicles, route optimization (Google Maps).
- Agriculture** — Crop yield prediction, precision farming.
- Education** — Personalized learning, dropout prediction.

(b) Five Sources of Data:

Source	Example	Challenges
Time Series	Stock prices, weather data	Handling seasonality, trends, missing timestamps
Transactional	Bank records, e-commerce orders	High volume, real-time processing, consistency
Biological	Genomic data, patient records	Privacy (HIPAA), complex structures, large size
Spatial	GPS data, satellite imagery	Coordinate systems, large files, real-time needs
Social Network	Tweets, posts, comments	Unstructured text, noise, bots, ethics, API limits

Q13. Write a complete Python program to analyse a dataset using Pandas, NumPy, and Matplotlib.

Scenario: File `library_books.csv` has: `Book_ID`, `Title`, `Author`, `Genre`, `Pages`, `Rating`, `Price`, `Copies_Sold`

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# 1. Read the dataset
```

```

df = pd.read_csv('library_books.csv')
print("Shape:", df.shape)
print("Columns:", df.columns.tolist())
print(df.head())

# 2. Basic statistics
print("\nMean Rating:", df['Rating'].mean())
print("Median Pages:", df['Pages'].median())
print("Most common Genre:", df['Genre'].mode()[0])

# 3. Missing values
print("\nMissing values:")
print(df.isnull().sum())
df['Rating'] = df['Rating'].fillna(df['Rating'].mean())
df['Price'] = df['Price'].fillna(df['Price'].median())

# 4. Filter top-rated books
top_books = df[df['Rating'] >= 4.5]
print(f"\nBooks with Rating >= 4.5: {len(top_books)}")
print(top_books[['Title', 'Rating', 'Copies_Sold']])

# 5. Group by Genre
genre_stats = df.groupby('Genre').agg({
    'Rating': 'mean',
    'Copies_Sold': 'sum',
    'Price': 'mean'
}).round(2)
print("\nGenre Statistics:")
print(genre_stats)

# 6. Bar chart – Average rating by genre
plt.figure(figsize=(10, 5))
genre_stats['Rating'].plot(kind='bar', color='teal', edgecolor='black')
plt.title('Average Rating by Genre')
plt.xlabel('Genre')
plt.ylabel('Average Rating')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# 7. Scatter plot – Pages vs Rating
plt.figure(figsize=(8, 5))
plt.scatter(df['Pages'], df['Rating'], alpha=0.6, color='purple')
plt.title('Pages vs Rating')
plt.xlabel('Number of Pages')
plt.ylabel('Rating')
plt.grid(True, alpha=0.3)
plt.show()

# 8. Save
df.to_csv('library_cleaned.csv', index=False)
print("Saved cleaned data!")

```

Q14. Explain Supervised, Unsupervised, Semi-Supervised, and Reinforcement Learning with examples.

(a) Supervised Learning (5 Marks):

- **Labelled data** (input-output pairs provided).
- Model learns the mapping from input → output.
- **Types:** Classification (predicting categories), Regression (predicting numbers).

- **Algorithms:** k-NN, Decision Tree, Linear Regression, SVM.
- **Example:** Email spam detection — model trained on emails labelled as spam/not spam.

(b) Unsupervised Learning (5 Marks):

- **Unlabelled data** (only inputs, no outputs).
- Model discovers hidden patterns or groupings.
- **Types:** Clustering, Association Rule Mining, Dimensionality Reduction.
- **Algorithms:** K-Means, DBSCAN, Apriori, PCA.
- **Example:** Customer segmentation — grouping users by purchasing behaviour without predefined categories.

(c) Semi-Supervised Learning (5 Marks):

- Combination of **small amount of labelled data + large amount of unlabelled data**.
- Model uses labelled data for supervised learning and leverages unlabelled data for better generalization.
- Useful when labelling data is expensive or time-consuming.
- **Algorithms:** Self-training, Label Propagation.
- **Example:** Medical image classification — only a few X-rays are labelled by doctors, but thousands of unlabelled X-rays are available.

(d) Reinforcement Learning (5 Marks):

- **Agent** learns by interacting with an **environment** and receiving **rewards/penalties**.
- No labelled data; model learns through trial and error.
- Goal: maximize cumulative reward.
- **Key Concepts:** Agent, State, Action, Reward, Policy.
- **Algorithms:** Q-Learning, Deep Q-Network (DQN), Policy Gradient.
- **Example:** AlphaGo — Google's AI learned to play Go by playing millions of games against itself, receiving rewards for winning.

Comparison Table:

Aspect	Supervised	Unsupervised	Semi-supervised	Reinforcement
Data	Labelled	Unlabelled	Mix	No data (environment)
Feedback	Direct (correct answer)	None	Partial	Reward/penalty
Goal	Predict	Discover patterns	Predict with less labels	Maximize reward
Example	Spam filter	Customer groups	Medical imaging	Game AI